

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 1 of 40

Topic: Pandas Series

Definition: Pandas Series is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 1: Create a Pandas Series.

Input

```
[10, 20, 30]
```

Program

```
import pandas as pd
s = pd.Series([10, 20, 30])
print(s)
```

Output

```
0 10
1 20
2 30
dtype: int64
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 2 of 40

Topic: DataFrame Creation

Definition: DataFrame Creation is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 2: Create a DataFrame from a dictionary.

Input

name and age columns

Program

```
import pandas as pd
df = pd.DataFrame({'name': ['A', 'B'], 'age': [20, 25]})
print(df)
```

Output

DataFrame with rows A/20 and B/25

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 3 of 40

Topic: Reading CSV Files

Definition: Reading CSV Files is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 3: Read a CSV file.

Input

```
sales.csv
```

Program

```
import pandas as pd
df = pd.read_csv('sales.csv')
print(df.head())
```

Output

```
First five rows of sales.csv
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 4 of 40

Topic: DataFrame head()

Definition: DataFrame head() is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 4: Display the first five rows.

Input

```
Existing DataFrame df
```

Program

```
print(df.head())
```

Output

```
First five rows
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 5 of 40

Topic: DataFrame tail()

Definition: DataFrame tail() is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 5: Display the last five rows.

Input

```
Existing DataFrame df
```

Program

```
print(df.tail())
```

Output

```
Last five rows
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 6 of 40

Topic: DataFrame Shape

Definition: DataFrame Shape is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 6: Find DataFrame shape.

Input

```
Existing DataFrame df
```

Program

```
print(df.shape)
```

Output

```
(rows, columns)
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 7 of 40

Topic: Column Names

Definition: Column Names is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 7: List column names.

Input

```
Existing DataFrame df
```

Program

```
print(df.columns.tolist())
```

Output

```
List of column names
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 8 of 40

Topic: Selecting One Column

Definition: Selecting One Column is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 8: Select one column.

Input

Column sales

Program

```
print(df['sales'])
```

Output

Sales column

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 9 of 40

Topic: Selecting Multiple Columns

Definition: Selecting Multiple Columns is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 9: Select multiple columns.

Input

name and sales

Program

```
print(df[['name', 'sales']])
```

Output

Selected columns

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 10 of 40

Topic: Filtering Rows

Definition: Filtering Rows is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 10: Filter rows by a condition.

Input

```
sales > 1000
```

Program

```
print(df[df['sales'] > 1000])
```

Output

```
Rows where sales > 1000
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 11 of 40

Topic: Sorting DataFrame

Definition: Sorting DataFrame is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 11: Sort by a column.

Input

```
sales ascending
```

Program

```
print(df.sort_values('sales'))
```

Output

```
DataFrame sorted by sales
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 12 of 40

Topic: Descending Sort

Definition: Descending Sort is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 12: Sort descending.

Input

```
sales descending
```

Program

```
print(df.sort_values('sales', ascending=False))
```

Output

```
DataFrame sorted descending
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 13 of 40

Topic: Missing Values

Definition: Missing Values is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 13: Find missing values.

Input

```
Existing DataFrame df
```

Program

```
print(df.isna().sum())
```

Output

```
Missing-value count per column
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 14 of 40

Topic: Missing Values

Definition: Missing Values is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 14: Fill missing values with zero.

Input

```
sales column
```

Program

```
df['sales'] = df['sales'].fillna(0)
print(df['sales'])
```

Output

```
Missing sales replaced by 0
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 15 of 40

Topic: Missing Values

Definition: Missing Values is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 15: Drop rows containing missing values.

Input

```
Existing DataFrame df
```

Program

```
df2 = df.dropna()  
print(df2)
```

Output

```
Rows with missing values removed
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 16 of 40

Topic: Duplicate Rows

Definition: Duplicate Rows is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 16: Remove duplicate rows.

Input

```
Existing DataFrame df
```

Program

```
df2 = df.drop_duplicates()  
print(df2)
```

Output

```
Duplicate rows removed
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 17 of 40

Topic: Renaming Columns

Definition: Renaming Columns is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 17: Rename a column.

Input

```
sales to revenue
```

Program

```
df = df.rename(columns={'sales': 'revenue'})  
print(df.columns)
```

Output

```
Column renamed to revenue
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 18 of 40

Topic: Changing Data Type

Definition: Changing Data Type is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 18: Change a column data type.

Input

```
age to integer
```

Program

```
df['age'] = df['age'].astype(int)
print(df.dtypes['age'])
```

Output

```
int64
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 19 of 40

Topic: Calculated Column

Definition: Calculated Column is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 19: Create a calculated column.

Input

```
price * quantity
```

Program

```
df['total'] = df['price'] * df['quantity']  
print(df['total'])
```

Output

```
Calculated total column
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 20 of 40

Topic: Column Mean

Definition: Column Mean is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 20: Find the mean of a column.

Input

```
sales column
```

Program

```
print(df['sales'].mean())
```

Output

```
Mean sales value
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 21 of 40

Topic: Group Totals

Definition: Group Totals is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 21: Find group totals.

Input

```
category and sales
```

Program

```
print(df.groupby('category')['sales'].sum())
```

Output

```
Total sales by category
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 22 of 40

Topic: Category Counts

Definition: Category Counts is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 22: Count records by category.

Input

category column

Program

```
print(df['category'].value_counts())
```

Output

Frequency per category

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 23 of 40

Topic: Group Maximum

Definition: Group Maximum is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 23: Find maximum value per group.

Input

category and sales

Program

```
print(df.groupby('category')['sales'].max())
```

Output

Maximum sales by category

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Question 24 of 40

Topic: Merging DataFrames

Definition: Merging DataFrames is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 24: Merge two DataFrames.

Input

```
customer_id as key
```

Program

```
merged = pd.merge(customers, orders, on='customer_id')  
print(merged)
```

Output

```
Merged DataFrame
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 25 of 40

Topic: Concatenating DataFrames

Definition: Concatenating DataFrames is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 25: Concatenate two DataFrames.

Input

df1 and df2

Program

```
result = pd.concat([df1, df2], ignore_index=True)
print(result)
```

Output

Rows combined

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 26 of 40

Topic: Pivot Table

Definition: Pivot Table is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 26: Create a pivot table.

Input

```
category, region, sales
```

Program

```
pivot = df.pivot_table(values='sales', index='category',  
columns='region', aggfunc='sum')  
print(pivot)
```

Output

```
Sales summary by category and region
```

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Question 27 of 40

Topic: Applying Functions

Definition: Applying Functions is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 27: Apply a function to a column.

Input

```
double sales
```

Program

```
df['double'] = df['sales'].apply(lambda x: x * 2)
print(df['double'])
```

Output

```
Sales values doubled
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 28 of 40

Topic: Year Extraction

Definition: Year Extraction is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 28: Extract year from a date.

Input

```
2026-08-12
```

Program

```
df['date'] = pd.to_datetime(df['date'])  
print(df['date'].dt.year)
```

Output

```
2026
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 29 of 40

Topic: Multiple Conditions

Definition: Multiple Conditions is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 29: Filter using multiple conditions.

Input

```
sales > 100 and region = South
```

Program

```
result = df[(df['sales'] > 100) & (df['region'] == 'South')]  
print(result)
```

Output

```
Matching rows
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 30 of 40

Topic: Unique Values

Definition: Unique Values is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 30: Find unique values.

Input

```
region column
```

Program

```
print(df['region'].unique())
```

Output

```
Unique regions
```

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Question 31 of 40

Topic: Unique Values

Definition: Unique Values is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 31: Find number of unique values.

Input

```
region column
```

Program

```
print(df['region'].nunique())
```

Output

```
Number of unique regions
```

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Question 32 of 40

Topic: Replacing Values

Definition: Replacing Values is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 32: Replace a value.

Input

```
'M' with 'Male'
```

Program

```
df['gender'] = df['gender'].replace({'M': 'Male'})  
print(df['gender'])
```

Output

```
M replaced with Male
```

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Question 33 of 40

Topic: Setting an Index

Definition: Setting an Index is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 33: Set an index.

Input

```
customer_id
```

Program

```
df = df.set_index('customer_id')  
print(df.index)
```

Output

```
customer_id becomes index
```

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Question 34 of 40

Topic: Setting an Index

Definition: Setting an Index is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 34: Reset an index.

Input

Existing indexed DataFrame

Program

```
df = df.reset_index()  
print(df.head())
```

Output

Default integer index restored

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 35 of 40

Topic: CSV Export

Definition: CSV Export is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 35: Export DataFrame to CSV.

Input

```
output.csv
```

Program

```
df.to_csv('output.csv', index=False)
```

Output

```
output.csv created
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 36 of 40

Topic: Duplicate Value Detection

Definition: Duplicate Value Detection is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 36: Find duplicated values in a column.

Input

```
customer_id
```

Program

```
print(df['customer_id'].duplicated())
```

Output

```
Boolean duplicate flags
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 37 of 40

Topic: Removing Columns

Definition: Removing Columns is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 37: Remove columns.

Input

```
temporary column
```

Program

```
df = df.drop(columns=['temporary'])  
print(df.columns)
```

Output

```
temporary column removed
```

TOP 40 PANDAS INTERVIEW QUESTIONS

Question 38 of 40

Topic: Summary Statistics

Definition: Summary Statistics is a Pandas technique used to work with tabular data. It helps prepare, transform, summarize, or analyze DataFrame information efficiently.

Question 38: Get summary statistics.

Input

```
Numeric DataFrame
```

Program

```
print(df.describe())
```

Output

```
Count, mean, std, min, quartiles and max
```